3.4.3 In photosynthesis, energy is transferred to ATP in the light-dependent reaction and the ATP is utilised in the light-independent reaction.

Photosynthesis	The light-independent and light-dependent reactions in a typical C3 plant.
Light-dependent reaction	The light-dependent reaction in such detail as to show that • light energy excites electrons in chlorophyll • energy from these excited electrons generates ATP and reduced NADP • the production of ATP involves electron transfer associated with the electron transfer chain in chloroplast membranes • photolysis of water produces protons, electrons and oxygen.
Light-independent reaction	The light-independent reaction in such detail as to show that • carbon dioxide is accepted by ribulose biphosphate (RuBP) to form two molecules of glycerate 3-phosphate (GP) • ATP and reduced NADP are required for the reduction of GP to triose phosphate • RuBP is regenerated in the Calvin cycle • Triose phosphate is converted to useful organic substances.
Limiting factors	The principle of limiting factors as applied to the effects of temperature, carbon dioxide concentration and light intensity on the rate of photosynthesis. **Candidates should be able to** explain how growers apply a knowledge of limiting factors in enhancing temperature, carbon dioxide concentration and light intensity in commercial glasshouses. They should also be able to evaluate such applications using appropriate data.